

Design Document — Personal Homepage

Author: Ayush Nimeshkumar Patel **Course:** [CS5610 Web Development](#), Northeastern University
Project: Project 1 — Your personal home page **Live site:** <https://ayushp2207.github.io/ayush-homepage/> **Repository:** <https://github.com/ayushp2207/ayush-homepage>

1. Project description

What this is

A static personal homepage built with vanilla HTML5, CSS3 and ES6 modules. No backend, no framework, no component library, no build step. Bootstrap 5 is used for its grid only; every other style is hand-written.

The problem it solves

A résumé is a list of claims. "Kubernetes, PyTorch, LangChain" tells a reader that I have touched those things, but not *where*, *when*, or *how seriously*. Recruiters skim for keywords; engineers want evidence. The two audiences need the same facts presented at different depths, and a flat list of bullet points serves neither well.

So the organising idea of this site is **evidence, not adjectives**. Every skill claim is wired to the specific role, project or paper where I used it, and the site makes that wiring browsable rather than asking the reader to reconstruct it from prose.

Scope

Three pages:

Page	URL	Purpose
Home	<code>index.html</code>	Identity, the skill constellation, experience, education, toolkit
Projects	<code>projects.html</code>	Longer write-ups of projects and publications
Built with AI	<code>built-with-ai.html</code>	The assignment's AI-generated page, used as a full GenAI disclosure

Non-goals

- No blog or CMS. Content lives in one JavaScript module and is edited by hand.
 - No contact form, because that would need a backend.
 - No dark/light toggle. The site commits to one dark palette and does it well.
-

2. User personas

Persona 1 — Priya, technical recruiter

- **Role:** University recruiter at a cloud infrastructure company.
- **Context:** Screening ~120 candidates for summer internships. Roughly 40 seconds per profile on the first pass, on a laptop with a dozen tabs open.
- **Goals:** Confirm graduation date, work authorisation timing, and whether the candidate has shipped anything at production scale.
- **Frustrations:** Portfolios that open with a full-screen animation and bury the graduation date three scrolls down. Having to open a PDF to learn anything concrete.
- **What she needs from this site:** Name, current role, graduation date and availability visible without scrolling. One click to email.

Persona 2 — Daniel, engineering hiring manager

- **Role:** Senior engineer who will run the technical interview.
- **Context:** Already read the résumé and is deciding whether to spend an hour on a phone screen. Reads carefully, and is sceptical of inflated claims.
- **Goals:** Find out whether the candidate's "multi-agent systems" experience is a tutorial project or real work; see actual code.
- **Frustrations:** Skill bars showing "Python 90%". Project cards with a title, a logo and no explanation of what was actually built or how hard it was.
- **What he needs from this site:** The ability to pick a technology and immediately see every place it was used, with enough detail to form a question. A link to source.

Persona 3 — Meera, fellow student and peer reviewer

- **Role:** Classmate assigned to review this submission.

- **Context:** On a phone, between classes, working through a rubric.
 - **Goals:** Check the rubric items — semantic markup, responsive layout, alt text, original JS, MIT license, README quality.
 - **Frustrations:** Sites that only work at desktop width. Creative additions that turn out to be a copied library demo.
 - **What she needs from this site:** A layout that holds up at 390px, a clearly identified creative addition, and an honest statement of what was generated.
-

3. User stories

Written as stories with acceptance criteria, so each one is testable.

Epic A — Establish identity fast

A1. *As Priya, screening quickly, I want the candidate's name, current role and availability without scrolling, so I can triage in seconds.*

- Given a 1440×900 desktop viewport
- When the homepage loads
- Then the name, a one-sentence summary, and "Available ... starting Summer 2027" are all visible above the fold
- And an email link is reachable in one click

A2. *As Priya, I want to know where the candidate studies and when they graduate, so I can match them to a requisition.*

- Given the homepage
- When I reach the Education section
- Then each degree shows institution, degree name, expected date and location

Epic B — Turn skill claims into evidence

B1. *As Daniel, I want to select a technology and see every place it was used, so I can tell real experience from a weekend tutorial.*

- Given the skill constellation
- When I select the node "Kubernetes"

- Then a detail panel lists every role, project and publication using it
- And it states how many places that is
- And unrelated nodes visibly dim

B2. *As Daniel, I want to go the other direction — pick a role and see what it was built with.*

- Given the constellation
- When I select the node "Agentic 5G Spectrum Testbed"
- Then the panel shows that role's summary and the skills it used

B3. *As Daniel, I want to clear a selection without reloading.*

- Given a selected node
- When I select the same node again, click empty canvas, or press "Clear selection"
- Then the filter clears and the panel returns to its prompt

B4. *As Meera on a phone, I want the same filtering without a mouse.*

- Given a touch device or keyboard-only navigation
- When I Tab to the node list and press Enter
- Then that node is selected, aria-pressed becomes true, and the same detail panel updates
- So the canvas is never the only route to the information

Epic C — Depth on demand

C1. *As Daniel, I want longer write-ups than a résumé bullet, so I can prepare questions.*

- Given the Projects page
- Then each entry has a summary plus specific detail points
- And publications are listed with venue and status

Epic D — Work everywhere, for everyone

D1. *As Meera on a 390px phone, I want a usable layout.*

- Given a 390px viewport
- Then the page never scrolls horizontally
- And the navigation collapses behind a labelled Menu button
- And the constellation still renders and responds

D2. *As a keyboard user, I want to skip repeated navigation.*

- Given focus at the top of the document
- When I press Tab once
- Then a "Skip to main content" link appears and is focusable

D3. *As a screen reader user, I want the graph not to be a dead end.*

- Given the canvas
- Then it exposes `role="img"` and a description saying the same nodes exist as buttons below

D4. *As someone with vestibular sensitivity, I don't want a graph animating at me.*

- Given `prefers-reduced-motion: reduce`
- Then the layout is computed without animating and painted once

Epic E — Academic honesty

E1. *As the grader, I want to know exactly which parts were AI-generated.*

- Given the "Built with AI" page
- Then it names the model and version, lists what was asked, separates generated from hand-authored content, and records where the model was wrong

4. Information architecture

```
index.html ..... Home
├── Hero ..... name, one-line pitch, contact, availability
├── Skill constellation the creative addition
├── Experience ..... reverse-chronological timeline (4 roles)
├── Education ..... 2 degrees
├── Toolkit ..... grouped skill lists
projects.html ..... Projects (3) + Publications (2) + contact
built-with-ai.html ..... GenAI disclosure (the AI-generated page)
```

All three pages share one persistent header and footer. Content is defined once in `js/data.js`; the constellation and the rendered sections both read from it, so a role is never described in two places.

5. Design mockups

Wireframes produced before implementation. Final screenshots follow in section 8.

5.1 Home — desktop (≥ 992px)

[AP] Ayush Patel

HomeProjectsBuilt with AIGitHub

sticky

BOSTON, MA

Ayush Nimeshkumar Patel

<- h1, clamp()

MS CS at Northeastern, currently building control-plane services at AWS. Multi-agent AI systems and the infrastructure that makes them reliable.

(See my work) (email) (LinkedIn) (GitHub)

<- pill links

| Available for full-time roles starting Summer 2027

<- accent rule

SKILL CONSTELLATION

A résumé tells you someone knows Kubernetes. It rarely tells you where they used it.

Databricks

DNS Agents

Python

Pit Wall

Rand-PC

MCP

Pibit

5G Testbed

Skill

Experience

Project

Publication

PICK A NODE

(AWS)(Computer Vision)(Databricks)

(Deep RL)(DynamoDB)(Java)(Python)

(Kubernetes)(LangChain)(MCP) ...

(Clear selection)

7 cols

Python

7 PLACES I HAVE USED IT

- DNS Anomaly Detection

- 5G Spectrum Testbed

- Pit Wall

5 cols

EXPERIENCE

SEP 2026 – PRESENT

Software Development Engineer Intern

Amazon Web Services · Boston, MA

Designing S3 endpoint configuration for Transfer Family...

JUN 2026 – AUG 2026

AI Engineer Intern ...

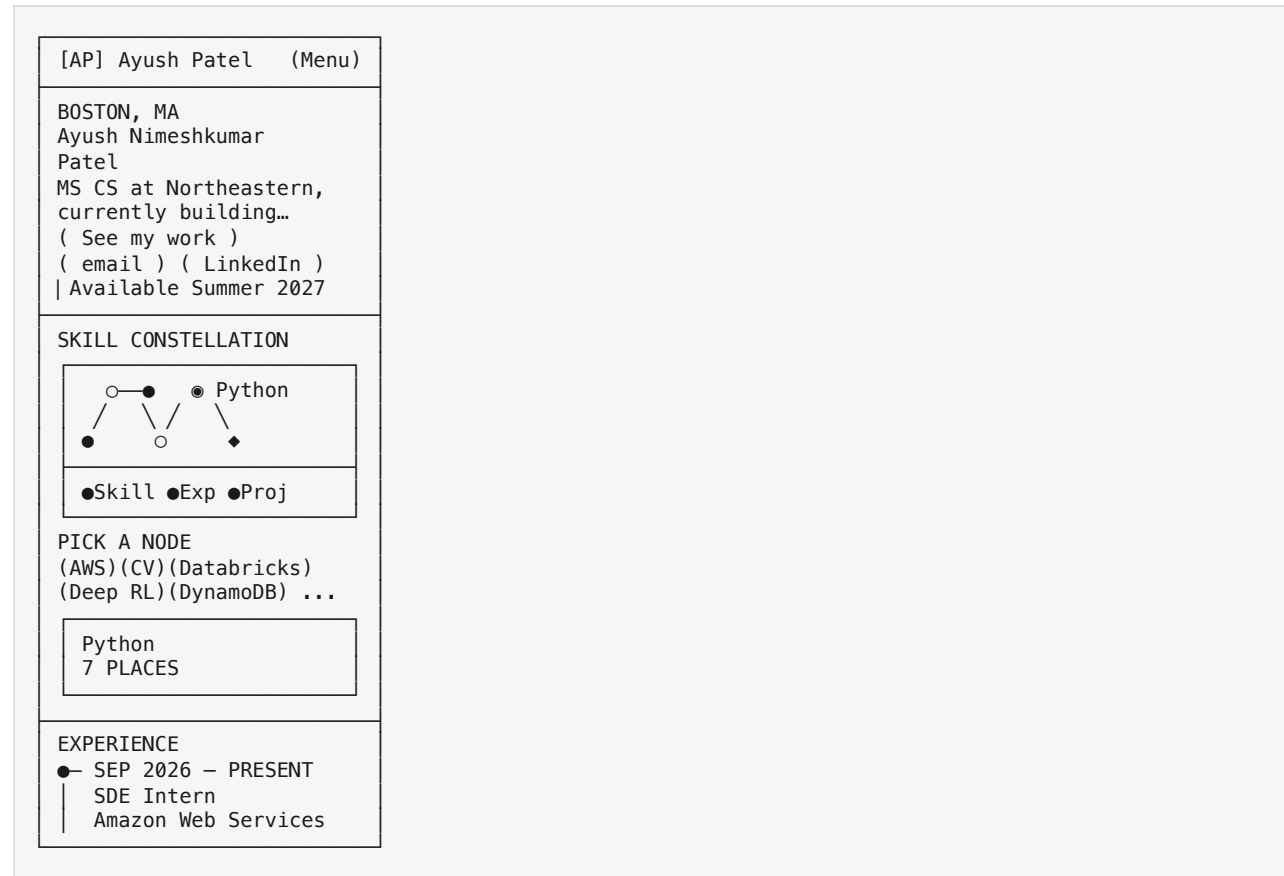
EDUCATION

TOOLKIT

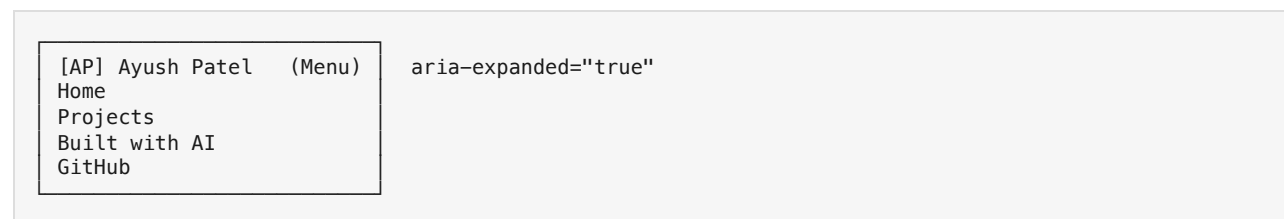
© 2026 Ayush Patel · Built for CS5610 · Home Projects Source

5.2 Home — mobile (< 768px)

Single column throughout. Nav collapses; the constellation keeps its full interaction but shortens to 340px tall; the detail panel stacks under the chips.

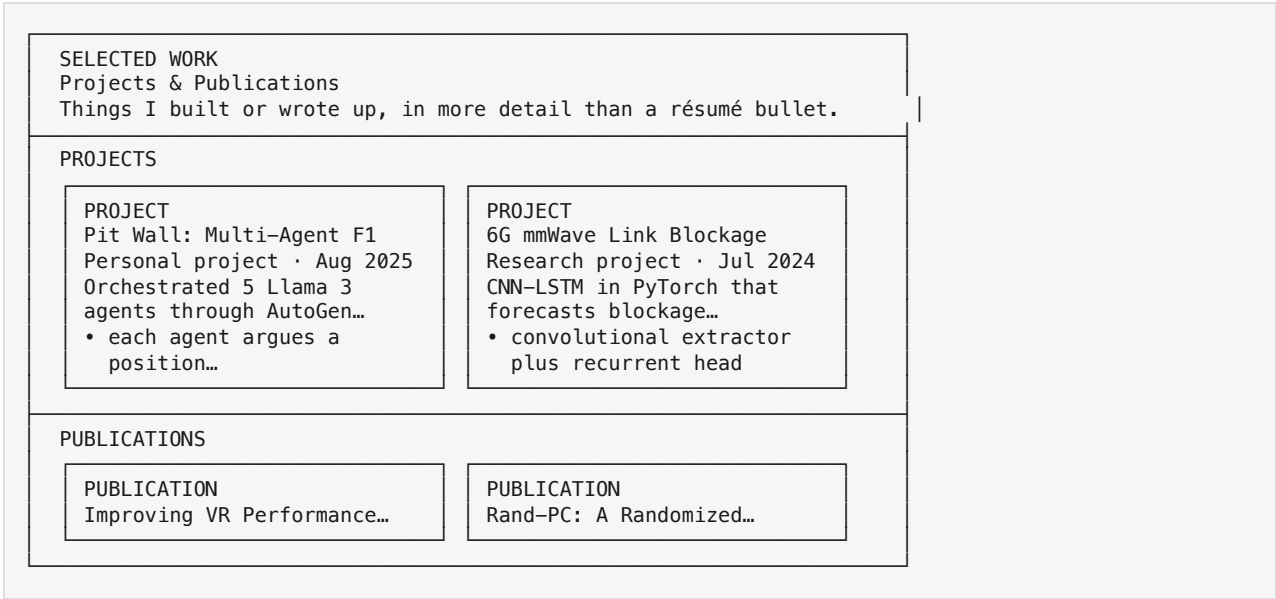


Nav open state:

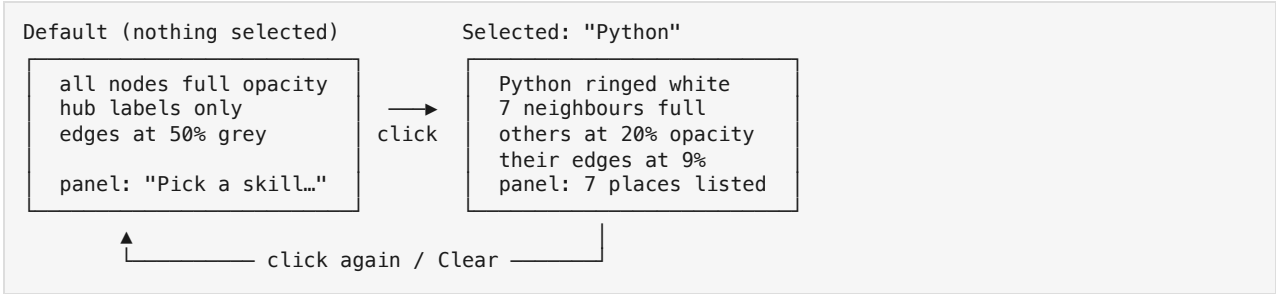


5.3 Projects — desktop

Two-column card grid (col-12 col-lg-6), collapsing to one column on mobile.



5.4 Constellation interaction states



6. Design system

Palette

Dark, low-chroma background so the four node colours carry the meaning.

Token	Value	Use
--ink	#0b1020	Page background
--surface	#111827	Cards, nav, graph stage
--surface-raised	#1b2337	Pills and chips
--line	#2b3550	All borders and rules

Token	Value	Use
--text	#e2e8f0	Body text
--text-muted	#94a3b8	Secondary text
--accent	#5eead4	Links, skill nodes
--accent-alt	#818cf8	Experience nodes
--accent-warm	#f472b6	Project nodes
--accent-gold	#fbbf24	Publication nodes

Node colour is the same in the graph, the legend, and the card accents, so the encoding is learned once.

Type and spacing

System font stack, so there is no web-font round trip. The h1 uses `clamp(2.1rem, 6vw, 3.4rem)` to scale without breakpoints. Body copy is capped at 58–70ch for readability. Spacing is a six-step scale ($0.25/0.5/1/1.5/2.5/4\text{rem}$) exposed as custom properties, so vertical rhythm stays consistent.

CSS conventions

- Every element is targeted by **class**, never by id, and never by bare tag beyond base resets.
- **Zero !important** in the stylesheet (verified: `grep -c '!important' → 0`).
- Layout uses Bootstrap's grid plus CSS flexbox and grid. One deliberate override: `.page-wrap .row` zeroes Bootstrap's negative gutters, because `.page-wrap` is not a Bootstrap `.container` and the negative margins otherwise cause horizontal overflow on mobile.

Accessibility decisions

Concern	Decision
Keyboard access to the graph	Every node is also a real <code><button></code> with <code>aria-pressed</code>
Screen readers	Canvas is <code>role="img"</code> with a label pointing at the button list
Skip navigation	<code>.skip-link</code> , off-screen until focused
Focus visibility	One shared <code>:focus-visible</code> outline, 3px, offset 3px
Motion sensitivity	<code>prefers-reduced-motion</code> settles the layout without animating

Concern	Decision
Semantic controls	Buttons are <code><button></code> , links are <code><a></code> — no clickable <code><div></code>
Images	Every <code></code> carries a real <code>alt</code>

7. The creative addition: Skill Constellation

Why this and not a honeycomb grid

The assignment's own example is a honeycomb image grid, so that is the one thing guaranteed to appear in other submissions. More importantly, a decorative grid does not answer the question a hiring manager actually has. The constellation is chosen because it is the *only* element on the site that does something a static résumé cannot: it inverts the relationship between skill and evidence on demand.

How it works

A force-directed graph on a `<canvas>`, using the Fruchterman–Reingold formulation:

- **Repulsion** between every pair of nodes: $f = k^2 / d$
- **Attraction** along every edge: $f = d^2 / k$
- **Ideal distance** $k = 0.62 \cdot \sqrt{(\text{area} / \text{nodeCount})}$ — derived from the canvas size and node count, so the same code lays out correctly at 396px and 1098px with no breakpoint-specific constants
- **Cooling**: a temperature caps how far any node may move per step and decays by 2.5% per frame, so the graph converges instead of oscillating
- **Weak centring** keeps disconnected components from drifting off-frame

23 nodes (14 skills + 9 work items) and 26 edges. Animation halts once total movement falls below a threshold, so it does not burn CPU idling.

Labels are drawn in a second pass, in priority order, with rectangle-collision detection: a label that would overlap one already drawn is skipped. Without this the graph is unreadable — the first implementation drew all 23 labels and they piled on top of each other.

Honest limitations

- The layout is deterministic by seeding positions on two rings, but it is still a physical simulation, so a resize past 40px reseeds and re-settles.
- Disconnected clusters drift apart. The AWS role sits on its own because Java, DynamoDB and AWS are not shared with any other entry. That is truthful, and arguably the most informative thing the graph says.

8. Final implementation

Home — hero

 Ayush Patel

Home Projects Built with AI GitHub

BOSTON, MA

Ayush Nimeshkumar Patel

MS Computer Science student at Northeastern, currently building control-plane services at AWS. I spend most of my time on multi-agent AI systems, the infrastructure that makes them reliable, and the networks they run over.

[See my work](#)ayushnpatelworks@gmail.com[LinkedIn](#)[GitHub](#)

Available for full-time software engineering roles starting Summer 2027.



At the AT&T Innovation Center R&D Labs, Middletown NJ, during my Network CTO internship.

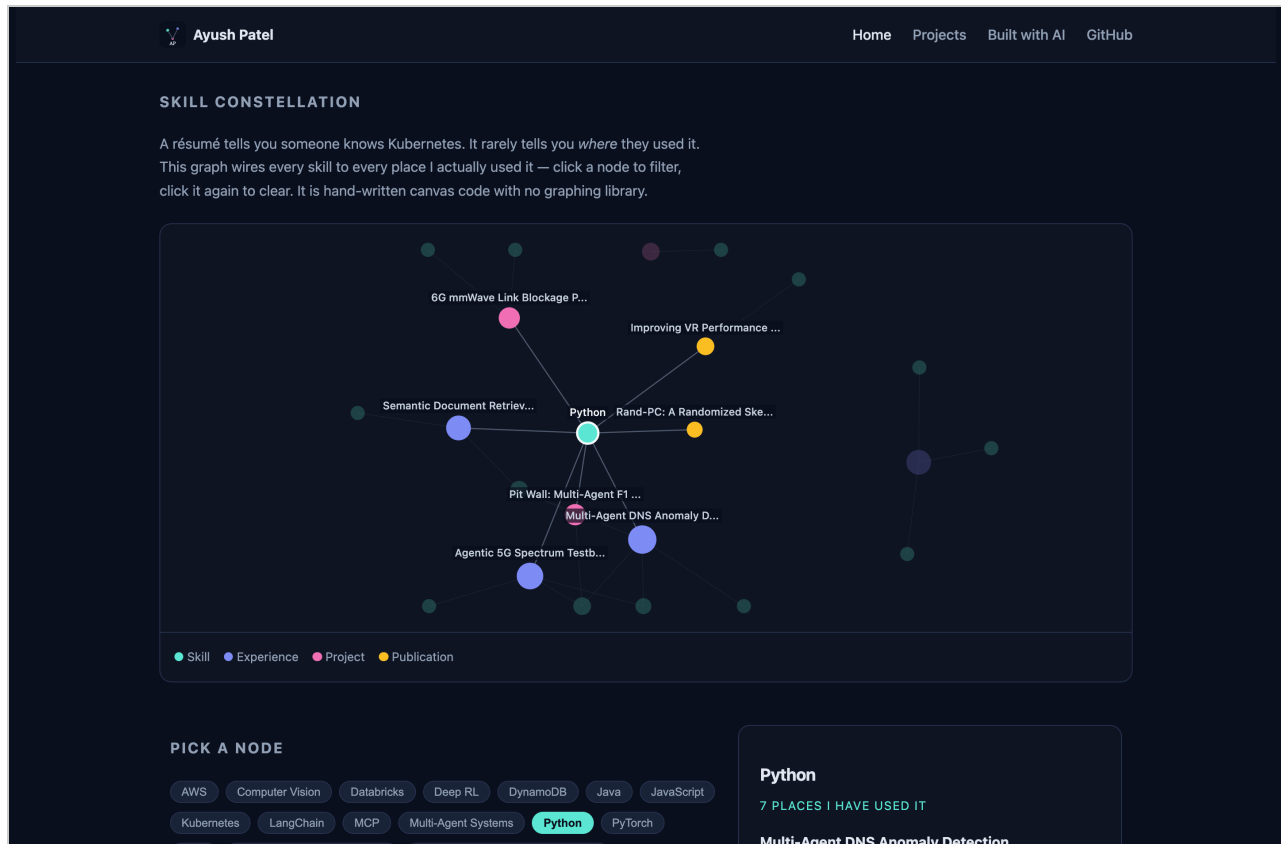
SKILL CONSTELLATION

A résumé tells you someone knows Kubernetes. It rarely tells you *where* they used it. This graph wires every skill to every place I actually used it — click a node to filter, click it again to clear. It is hand-written canvas code with no graphing library.

This Homepage



Skill constellation with "Python" selected



Python is ringed; its 7 connected work items stay at full opacity while the rest of the graph drops to 20%. The detail panel lists each one.

Mobile



Ayush Patel

Menu

BOSTON, MA

Ayush Nimeshkumar Patel

MS Computer Science student at Northeastern, currently building control-plane services at AWS. I spend most of my time on multi-agent AI systems, the infrastructure that makes them reliable, and the networks they run over.

[See my work](#)

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9. Verification

Check	Method	Result
W3C validity	validator.w3.org/nu on all 3 pages	0 errors
ESLint (class config)	npx eslint .	0 errors
Prettier	npx prettier --check	clean
Console errors	Headless Chrome, all 3 pages	none
Images have alt	DOM audit	0 missing
Horizontal scroll at 390px	scrollWidth vs innerWidth	none
Keyboard filtering	Tab + Enter on node list	works
!important count	grep -c on the stylesheet	0

